Aladdin by BlackRock

Quick Start Guide

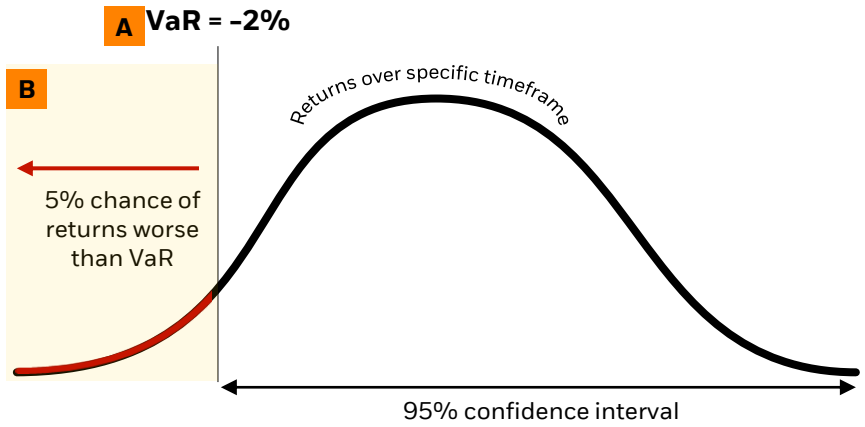
Authors:
Syed Imran Shah, *Vice President*, Aladdin Product
Andrea Woo, *Associate*, Aladdin Product

ALADDIN: VALUE AT RISK ANALYTICS

The Aladdin suite of models offers robust support for Value at Risk (VaR) analytics, which are essential for measuring and quantifying portfolio risk. Aladdin supports 3 different methods for computing VaR: **Analytical VaR, Historical VaR** and **Monte Carlo VaR**. These VaR analytics enable investors to gain a deeper understanding of their portfolio's risk profile and make informed decisions to mitigate potential losses.

What is VaR?

VaR is a statistical measure used to estimate the potential loss in value of a portfolio over a defined period for a given confidence interval. Essentially, it measures the worst expected loss under normal market conditions over a specific time frame at a certain confidence level.



A In this example, **VaR is estimated** to be -2% at 95% confidence interval.

B **Expected Shortfall**, also known as Conditional VaR**,** provides further insight by estimating the **average loss in those worst-case scenarios beyond the VaR.**

1 **Factor returns simulation**
Analyse potential portfolio losses by considering impact of various risk factors

2 **Security revaluation**
Recalculate value of securities under different risk scenarios to estimate potential losses

3 **Portfolio aggregation**
Combine individual security risks to assess overall portfolio risk

Other applications

Aside from quantifying risk in a simple term, estimating VaR has also several merits including –

LIMIT MONITORING
VaR can be used as a boundary to prevent excessive risk taking by triggering actions through alerts when the threshold has been breached

PORTFOLIO OPTMIZATION
Integrate VaR into portfolio optimization models to achieve best possible returns while keeping risk within acceptable limits

STRESS TESTING
Combine VaR with historical or hypothetical scenarios to evaluate how extreme shocks can impact portfolios

REGULATORY COMPLIANCE
Measure VaR to determine how much capital is required as a buffer against unexpected losses and report VaR as part of risk management processes

The risk models are integrated throughout Aladdin's interactive analytics, portfolio analysis and construction application to offer a comprehensive risk analysis and insights through tools such as Explore, PfC and Risk Radar[1]. Through Explore users can access portfolio, asset or factor level analytics in charting or tabular form.

1. Portfolio Construction & Risk Radar is currently only Available to Aladdin Enterprise users. To discuss further Risk Radar further, please contact your Aladdin client engagement team.

Covariance Matrix

The covariance matrix is essential to all portfolio analytics within the Aladdin Risk Model, enabling the estimation and forecasting of volatility, empirical betas, VaR, and stress testing. Aladdin provides users with flexibility in covariance matrix estimation, with the risk engine computing these matrices on-the-fly.

In the Aladdin Risk Model, the covariance matrix estimation is determined by **four key components**, each of which offers flexibility and customization to meet specific portfolio needs:

OBSERVATION WINDOW

Time period over which historical data is analyzed

A **larger observation window includes more historical data**, leading to a more stable covariance matrix, while a **shorter window makes it more sensitive to recent market** conditions.

WEIGHTING SCHEME

By assigning different weights to data points, this component affects how recent or past data influences the covariance matrix.

Higher weighting on recent data **emphasizes the latest market movements**, making the matrix more responsive to volatility shifts, while **equal weighting captures a balanced historical view.**

DATA FREQUENCY

How frequently data points are sampled e.g. daily, weekly, monthly

High-frequency data e.g., daily, makes the matrix **respond quicker** to market changes, while **lower-frequency data smooths out short-term variations** for a more stable risk profile.

OVERLAP VS NON OVERLAP

Choosing between overlapping data for more frequent return observations or non-overlapping data to reduce autocorrelation

Overlapping data increases robustness with more data points but can raise autocorrelation. **Non-overlapping data** reduces autocorrelation but may lead to more variability due to fewer data points.

The BlackRock Fundamental Equity Risk Model (BFRE) covariance matrix builds on standard methodology with three unique enhancements:

- **Distinct Half-Lives** for estimating correlations and volatilities.
- **Newey-West Asynchronicity Correction** instead of overlapping data.
- **Shrinkage Application** for improved stability and suitability in optimization.

Default settings

The Aladdin Risk Model includes default covariance matrix settings called "VaR Types." Users can select one of these or configure their own weighting scheme. Weekly matrices WKY and WKS use returns from Wednesday to Wednesday, with Tuesday used if Wednesday is a holiday. The MTC covariance matrix uses monthly returns defined by calendar month ends. DLY and WKS are similar, but WKS uses non-overlapping weekly returns, while DLY uses overlapping 5-day returns.

	Defaults			Configurable		
VaR Type	Half-life	Window	Overlap	Half-life	Window	Overlap
DLY	40 days	252 days	5	✓	✓	✓
WKS	8 weeks	104 weeks	1		✓	
WKY	26 weeks	104 weeks	1	✓	✓	
MTC	Equal weight	72 months	1	✓	✓	

Choosing covariance matrix parameters involves two main decisions, both related to the bias-variance tradeoff: **sampling frequency** and the **weight assigned to older data.**

Analytical VaR (AVaR)

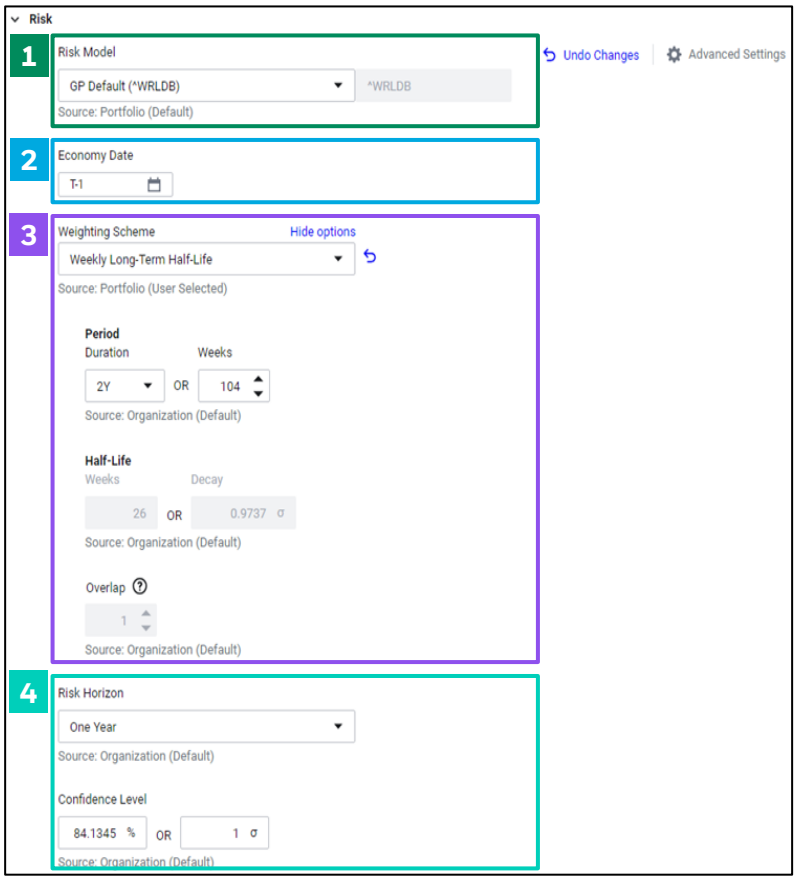
Analytical VaR uses a variance/covariance approach, assuming linear exposure to risk factors and a normal distribution of these factors. This method simplifies VaR calculation with statistical measures, making it widely used approach in financial risk management.

AVaR assumes that the security risk exposures are notionally weighted to the portfolio level, and it calculates portfolio volatility using the portfolio's risk factor exposures and the risk factor covariance matrix.

$$\sigma_P = \sqrt{\mathbf{e}^T \cdot \Omega \cdot \mathbf{e} + \mathbf{w}^T \cdot \Psi \cdot \mathbf{w}}$$

- **e** represents how sensitive the portfolio value is to changes in market risk factors.
- **Ω** is a matrix showing how these market risk factors are related to each other.
- **w** is a vector that contains the market value weights of each security in the portfolio.
- **Ψ** is a matrix that shows the risk specific to each security.

AVaR settings



1 Risk models determine the portfolio risk decomposition lens.

Risk models can be set up as organization or portfolio defaults. Users have the option to select a different model to ensure the appropriate risk lens and risk decompositions.

2 Economy date defines the risk date.

Setting the economy date locks the economic data, enabling users to analyze the current portfolio's exposures under the market conditions from a specific point in time.

3 Weighting scheme settings define the covariance matrix for the factor returns.

An exponentially weighted moving average on historical factor returns emphasizes recent returns, with the decay rate measured by a half-life metric, and overlapping observations enhancing statistical robustness.

4 Risk horizon and confidence level can be used to set the time-period for risk and certainty assessment.

Expected losses can be calculated across different horizons and confidence. Depending on the VaR model, users can pre-select horizons or custom the horizon. The confidence level can be set in % or standard deviation.

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Simple and fast VaR estimation
Estimation is streamlined by assuming a linear relationship between returns and risk factors

Does not require any extensive historical data
Use of variance-covariance approach to draw correlations between factors

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Does not capture nonlinearity
Overlooks nonlinear risks, potentially leading to an underestimation of overall risk

Not suited for skewed, fat tailed distributions
Financial markets deviate from normal distributions, AVaR underestimates large losses by failing to capture skewed or fat-tailed risks

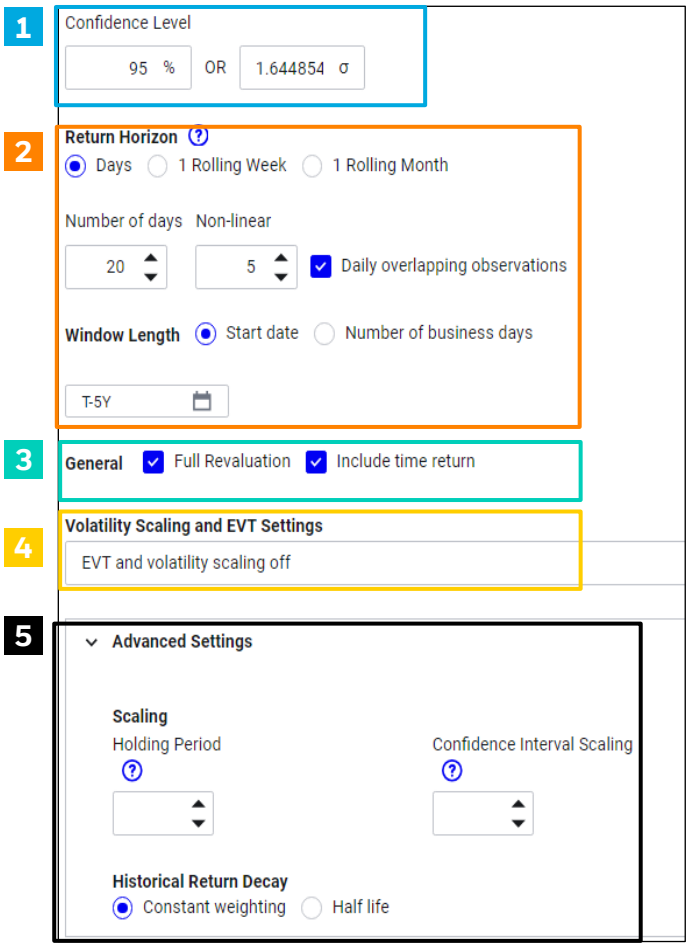
Historical VaR (HVaR)

HVaR estimates potential investment loss using historical data without assuming linear risk exposure or specific asset return distribution. Instead, it relies on actual historical returns.

Standard HVaR — **Enhanced HVaR**

Generate historical returns	Security Revaluation	Calculation of tail risk	Further calibration
Obtain samples of risk factor returns for the given time horizon	Each security are repriced using the sampled risk factor returns	Tail risk are computed using empirical distribution, allowing estimation of expected shortfall and tail analytics	Volatility scaling is used to create a VaR estimate that is more responsive to current market conditions Idiosyncratic risks captured using a parametric approach Extreme Value Theory and Kernel based method for extreme quantile estimation

HVaR settings



1. **Confidence interval** can be set in % or standard deviations
2. The **Return Horizon** works in conjunction with the **Window Length** to set the number of return periods used to calculate HVaR.
   - Day - uses a daily return
   - 1 Rolling Week / Month– the analysis date less one week / month is used as the first return observation

   Number of days define the return horizon used to estimate risk for assets with linear exposures whereas **non-linear** defines the horizon for assets with nonlinear exposures
   Daily overlapping observations[1] can be enabled/disabled.
3. **Full Revaluation** will revaluate all assets in the portfolio. **Time return can be included** in VaR estimation.
4. **Volatility scaling** enables rescaling of historical factor returns by ratio of current factor volatility. **EVT settings** refine empirical quantile estimation at all probability levels and can be applied at the portfolio level, factor level, or turned off.
5. **Holding period** scales risk estimates from shorter return horizons to longer horizons when data is limited. The units used for holding period is **days**[2] . For instance, if you want to scale to a yearly return horizon, the risk estimates will use the square root of the value in the holding period (in this case, 252) over the return horizon as a scaler (sqrt of 252/20) on the risk estimates from the original return horizon.
   Confidence interval scaling scales observations from smaller to bigger confidence **intervals**[3] .
   Historical return decay defines how returns are weighted in the computation.

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Incorporates non-linearity
Uses historical data that include complex, non-linear relationships between risk factors and returns

Include fat tails by using empirical distribution
Actual data captures extreme events, providing a more accurate representation of fat-tailed risks

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Requires significant history
Relies on extensive historical data for accurate risk estimation and analysis

Only reflects tail risk if there are tail events
Unable to reflect tail risk when dataset does not contain past instances of tail event

1. If you select this with a 5-day horizon, we will consider 5 return values (Mon-Fri, Tues-Mon). Without this, we will have 2 returns (Mon – Fri, Mon – Fri)
2. 1 year = 252 business days. To scale estimates from a 20-days return horizon to a year, the value entered should be the final scaled value i.e. 252 for this case.
3. Since tail events are rare and do not have many observations, risk estimates from 95% confidence interval can be scaled to 99% by entering 99.

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Monte Carlo VaR (MCVaR)

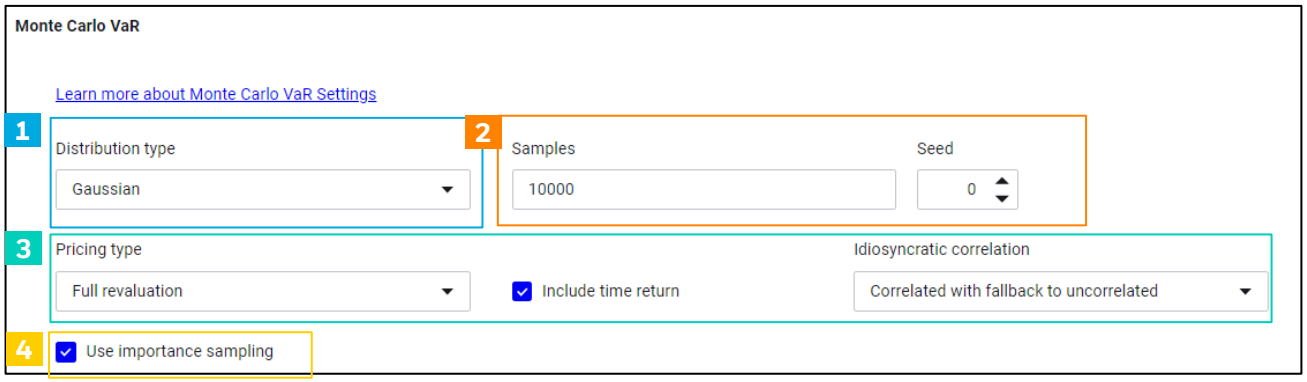
MCVaR involves simulating numerous possible returns for the portfolio's risk factors based on a chosen probability distribution. This approach is flexible and can handle complex risk profiles, but it is computationally intensive and sensitive to model assumptions

The specification of MCVaR can be viewed as the union of four sets of modeling assumptions

1. Joint probability distribution of risk factor returns
2. The simulation algorithm used to efficiently sample from the joint probability distribution;
3. The set of pricing functions used for pricing the portfolio securities given the simulated risk factors
4. The estimation of tail risk analytics, such as the contribution to VaR and ES, derived from the simulated security returns.

1. Risk factors are identified and used to model the joint distribution assuming Gaussian or T distribution
2. Scenarios are simulated to generate scenarios that represent possible outcomes of risk factors based on joint distribution
3. Security returns are repriced using full revaluation or parametric return pricing functions based on simulated risk factor scenarios
4. Importance sampling is used to accelerate algorithm convergence and oversample tail risk analytics at portfolio level

MCVaR settings



1. **Distribution type** determines joint distribution of risk factor returns.
2. Determine number of **samples** used. **Seed determines the starting point for random number generator** used to produce numbers that drive simulation.
3. **Pricing type** determines pricing functions used to price securities. **Idiosyncratic correlation** determines correlation assumptions of idiosyncratic factors.
4. **Importance sampling** accelerate convergence of algorithm and oversample regions of interest for tail risk analytics. This option should be enabled only if analytics at portfolio/benchmark/active level are in focus.

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Incorporates non-linearity
Simulates a wide range of potential market scenarios, capturing non-linear relationships between risk factors and return

Possible to include any set of distributional assumptions
Accommodates any distributional assumptions to model risk under various market conditions

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Computationally intensive
Multiple simulations of market scenarios results in the need for significant processing

Highly sensitive to model assumptions
Accuracy depends heavily on the chosen assumptions, making it highly sensitive to errors in input data or model parameters.